

LAB 7 – Inheritance

1. Simple Inheritance – Person and Student

```
class Person
{
    public string Name;
    public string Gender;
}
class Student : Person
{
    public int RollNo;
}

class Program
{
    static void Main(string[] args)
    {
        Student s1 = new Student();
        s1.Name = "Rahul Dawson";
        s1.Gender = "Male";
        s1.RollNo = 101;
        Console.WriteLine("Student details inheritance from person ");
        Console.WriteLine($"Name: {s1.Name}, Gender: {s1.Gender}, RollNo: {s1.RollNo}");
    }
}
```

```
yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab7$ dotnet run
Student details inheritance from person
Name: Rahul Dawson, Gender: Male, RollNo: 101
```

2. Simple Inheritance – Employee and Manager

```
class Employee
{
    public string Name;
    public int JoiningYear;
    public int Salary;
}
class Manager: Employee
{
    public int NoOfEmployees;
    public Employee[] Employees;
}
class Program
{
    static void Main(string[] args)
    {
        Manager m1 = new Manager();
        m1.NoOfEmployees = 3;
        m1.Employees = new Employee[m1.NoOfEmployees];
        m1.Employees[0] = new Employee
        {
            Name = "John",
            JoiningYear = 2010,
        }
    }
}
```

```

        Salary = 10000
    };
    m1.Employees[1] = new Employee
    {
        Name = "Jane",
        JoiningYear = 2011,
        Salary = 12000
    };
    m1.Employees[2] = new Employee
    {
        Name = "Jack",
        JoiningYear = 2012,
        Salary = 15000
    };
    Console.WriteLine("\nEmployee Details");
    foreach (Employee emp in m1.Employees)
    {
        Console.WriteLine("-----");
        Console.WriteLine("Name: " + emp.Name);
        Console.WriteLine("Joining Year: " + emp.JoiningYear);
        Console.WriteLine("Salary: " + emp.Salary);
    }
}
}

```

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Employee Details

Name: John

Joining Year: 2010

Salary: 10000

Name: Jane

Joining Year: 2011

Salary: 12000

Name: Jack

Joining Year: 2012

Salary: 15000

3. Multilevel inheritance – Vehicle → Car → ElectricCar

```

class Vehicle
{
    public string BrandName;
    public string ModelName;
}

class Car:Vehicle
{
    public int NumberOfDoors;
}

```

```

    }
    class ElectricCar:Car{
        public int BatteryCapacity;
    }

    class Program
    {
        static void Main(string[] args)
        {
            ElectricCar e1 = new ElectricCar();
            e1.BrandName = "Dipal";
            e1.ModelName = "V10";
            e1.BatteryCapacity = 450000;
            e1.NumberOfDoors = 4;
            Console.WriteLine("Vehicle Details ");
            Console.WriteLine($"BrandName: {e1.BrandName}, ModelName: {e1.ModelName},
            NumberOfDoors: {e1.NumberOfDoors}, BatteryCapacity: {e1.BatteryCapacity}");

        }
    }
}
yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab7$ dotnet run
Vehicle Details
BrandName: Dipal, ModelName: V10, NumberOfDoors: 4, BatteryCapacity: 450000

```

4. Base keyword – Calling base case method

```

class Animal
{
    public void ShowInfo()
    {
        Console.WriteLine("This is an animal");
    }
}
class Dog : Animal
{
    public void ShowInfo()
    {
        base.ShowInfo();
        Console.WriteLine("This is a dog");
    }
}
class Program
{
    static void Main(string[] args)
    {
        Dog d1 = new Dog();
        d1.ShowInfo();
    }
}
yanil@fedora:~/Desktop/6th_sem/DotNet/lab/Lab7$ dotnet run
This is an animal
This is a dog

```
